ONE-WAVE · PROPOSED BUILD · ANIME/MANGA TECHNICAL MANUAL

ANDROID

BODY & BRAIN

NO CONSCIOUSNESS CLAIM · PROPOSED ARCHITECTURE ONLY

CONTENTS

CHAPTER 0

WHY BUILD THIS

This book contains a hypothetical One-Wave brain and body architecture for an android. It is an active engineering proposal — not a claim that the design is conscious, biologically identical to a human, or already validated in any way.

"The build may deliberately test ideas from the Proposed One-Wave Consciousness book. A successful implementation would show the functions can be

engineered. It would not, by itself, prove subjective awareness."

THE CORE SAFETY RULE

NO MODULE GETS UNILATERAL IRREVERSIBLE CONTROL

- Dream Engine cannot directly execute irreversible action.
- Administrator cannot bypass the completed state report.
- M4 cannot independently commit.
- Every commitment must preserve a recoverable prior reference state.

CHAPTER 1

BIO-INSPIRED DESIGN

Biological systems are used here as **design inspiration** for engineering, not as a claim that the android literally is biological. Each principle below borrows a mathematical pattern from a real, cited mechanism and applies it to a control-system function.

DESIGN PRINCIPLE 1 — ATP SYSTEMS → ENERGY MANAGEMENT

E-526's real cellular energy balance ($dU/dt = P_{in} - P_{use} - P_{loss}$) and ATP kinetics map directly onto battery/power budget management: available power budget, an intermediate power-ready state, raw energy input, conversion and draw-down rates. The biology doesn't add new physics — it supplies a clean, already-analyzed equation set instead of deriving power management from scratch.

DESIGN PRINCIPLE 2 — NEURAL OSCILLATORS → TIMING AND COORDINATION

E-524's Kuramoto lattice maps onto multi-node timing coordination: each processing node's clock phase, synchronized by a shared signal strength.

> **REAL FINDING, CARRIED FORWARD HONESTLY**
>
> The synchronization order parameter settles to a fixed value under constant coupling — it does NOT naturally produce a build-up/relaxation cycle unless something depletes over time. For pure timing coordination, a settled R is the goal, not a cycle.

DESIGN PRINCIPLE 3 — CELLULAR GRADIENTS → TRANSPORT AND FLOW

E-526's gradient transport equation ($J = -D \cdot \nabla\mu$) maps onto signal/data flow between hierarchy levels in C-312, giving the existing 3:1 fan-in structure an actual mathematical reason for its direction, not just a stated rule.

DESIGN PRINCIPLE 4 — FEEDBACK LOOPS → REGULATION

G-710's regulated response ($Q_n = k_n \cdot F_n$) governs how the coordinator corrects a triggering unit — proportional, not overcorrecting. E-527's relaxation-cycle finding applies here specifically: if a corrective action draws down a finite resource, the system shows genuine build-up/relaxation behavior — worth designing for explicitly.

```
Power Input (S) -> ATP-style buffer (A) -> Available budget (U)
        |
        v
Sensor Units (L1) -- gradient flow --> Clusters (L2, 3:1)
        |
        v
Timing sync (Kuramoto) across all active nodes
        |
        v
Coordinator (L4) -- regulated response (Q=kF, targeted) --> correction
```

CHAPTER 2

FUNCTIONAL RUNTIME

All gate relationships are duplex — forward proposals and reverse constraints travel together.

```
Sensors / current field
<-> choice and candidate generation
<-> simulated movement
<-> M4 mirror and scale weighing
<-> state / risk / readiness report
<-> accepted commitment
<-> body movement and recursive feedback
```

PROPOSED COMPUTE MAPPING (JETSON-CLASS, ILLUSTRATIVE)

- **GPU:** Dream Engine candidate generation and parallel simulation
- **CPU:** Administrator — identity, permission, commitment logic
- **Persistent low-latency service:** M4 weighing and modulation
- **Protected storage:** Reference Ground
- **Recoverable workspace:** Working Ground

> **NOTE**
>
> This role mapping is a design proposal. The silicon is not literally transformed into biological brain regions.

COMMIT RULE

```
promote Working Ground to Reference Ground
only when
Gate 5 state report matches Gate 6 acceptance
```

CHAPTER 3

BODY INTERFACE & CONTROL

```
local sensor/actuator units
-> 3:1 local aggregation candidate
-> positional chains
-> central coordination
-> targeted command fan-out
```

The hierarchy is distributed rather than dictatorial. Local units hold local state and perform fast responses. Higher layers modulate, coordinate, and correct selected units rather than micromanaging every movement.

CONSCIOUSNESS BOUNDARY

This architecture is allowed to be informed by the One-Wave consciousness hypothesis. The build itself only demonstrates implemented function. Any claim about consciousness requires a separate test definition that does not yet exist.

CHAPTER 4

FUNCTIONAL RECURSIVE ARCHITECTURE

This chapter is the body-control synthesis specification — narrower than the full brain runtime, and built entirely on **corrected** groundings. Earlier drafts of this same architecture used "Five Mind" and "Six Mind" labels grounded in the wrong nodes; those groundings failed audit (see G-719) and are replaced here with the real math that actually held up.

```
Input -> Local Processing -> Communication -> System Regulation -> Output -> Feedba
```

COMMUNICATION LAYER — CORRECTED GROUNDING

CORRECTION ON RECORD

Δψ_i = ⟨ψ_j⟩ − ψ_i — A-111's own real neighbor-average term. This replaces an earlier grounding in B-213 Access Line, which was checked and ruled out: B-213 explicitly operates only between one coupled pair, not across many nodes. A-111's term already averages over multiple neighbors — the correct citation was sitting in the repo the whole time.

SYSTEM REGULATION LAYER — CORRECTED GROUNDING

CORRECTION ON RECORD

Q_n = k_n·F_n — G-710's real regulated-response formula. This replaces an earlier grounding in G-711 Gate 7, matched by shape only (a seventh element reviewing six others) with no actual content match — G-711 checks repository coherence, not system-wide regulation.

WHAT THIS RESTS ON

Layer	Citation
Local Processing	C-312
Communication	A-111 (corrected)
Regulation	G-710 (corrected)
Oscillation scales	E-507
Triangular/hexagonal movement lattice	D-408 / D-411 / D-412

MOVEMENT MEMORY BOUNDARY

The broader subconscious movement stack is defined in G-722 and Proposed Android Brain Chapters 3-4: sequence grammar schedules route families, Boltzmann dynamics reconstruct candidates, Hopfield dynamics settle learned motor patterns, local **-1(0)+1** choice executes, and binary oversight supplies only safety and permission.

GEOMETRY COUNT

Three mirrored planar axis pairs produce six directed neighbor routes around one center. The 3:1 planar count is not the 3:1 nerve aggregation ratio. D-411 keeps the domains separate; D-412 defines the simulation receipts.

HONEST GAP, NOT SMOOTHED OVER

Section 7's three-step transition (Move→Hold→Transition) is only a **partial** reduction of B-221's real six steps — the same honest gap already identified for the four-state consciousness version elsewhere in this project. Not yet resolved either way.